

QUIZ COMPETITION BUZZER For Eight Players/Teams

S.C. DWIVEDI

This is a comprehensive guide to a circuit that can be used for eight players/teams to participate in quiz competitions organised by schools or colleges. Normally, quiz circuits are designed with microcontrollers, but this simple circuit does not use any microcontroller. The heart of this circuit is a priority encoder, 74LS373. The author's prototype built on a breadboard is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram for the quiz competition buzzer. It is built around a step-down transformer (X1), a bridge rectifier (BR1), a 5V voltage regulator LM7805 (IC1), eight push-to-on switches (S1 through

S8), priority encoder IC 74LS373 (IC2), two NPN transistors 2N2219 (T1, T2), a piezo buzzer (PZ1), nine LEDs (LED1 through LED9), and a few other components.

The circuit operates on 5V, which is derived from the 230V AC mains. The 230V AC mains is stepped down to 9V AC by transformer X1, rectified by bridge rectifier BR1, filtered by capacitor C1, and regulated by LM7805. The priority encoder IC 74LS373 is used to transfer the logic state from data input pins D0 through D7 to the corresponding Q0 through Q7 outputs. All the data pins (D0 through D7) are pulled low by resistors R2 through R9, respectively, along with switches S1 through S8. The other ends of switches S1 through S8 are connected to +5V.

The heart of this quiz project is IC 74LS373. It consists of eight D-type latches/flip-flops with 3-state outputs.

The eight latches of the 74LS373 are transparent D-type latches, meaning that while the latch enable (LE) input is high, the Q outputs (Q0 through Q7) follow the data inputs (D0 through D7). When LE is low, the output is latched at the level of the data that was set up. This is the reason for controlling this LE (pin 11), where we have used transistor T1 along with resistor R10. This 74LS373 IC has a working voltage

PARTS LIST

Semiconductors:

- IC1 - LM7805, 5V voltage regulator
- IC2 - 74LS373 priority encoder
- T1, T2 - 2N2219 NPN transistors
- BR1 - 1A bridge rectifier
- LED1-LED9 - 5mm LEDs

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2-R10 - 5.6-kilo-ohm
- R11-R18 - 100-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- S1-S8 - Push-to-on switches
- X1 - 230V AC primary to 9V, 500mA secondary transformer

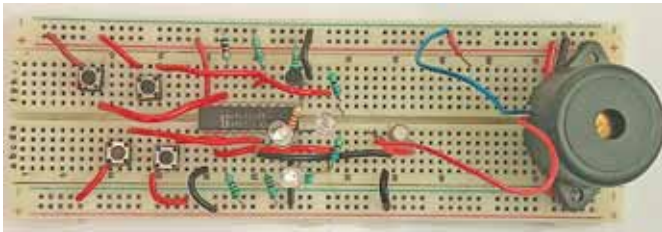


Fig. 1: Author's prototype on a breadboard

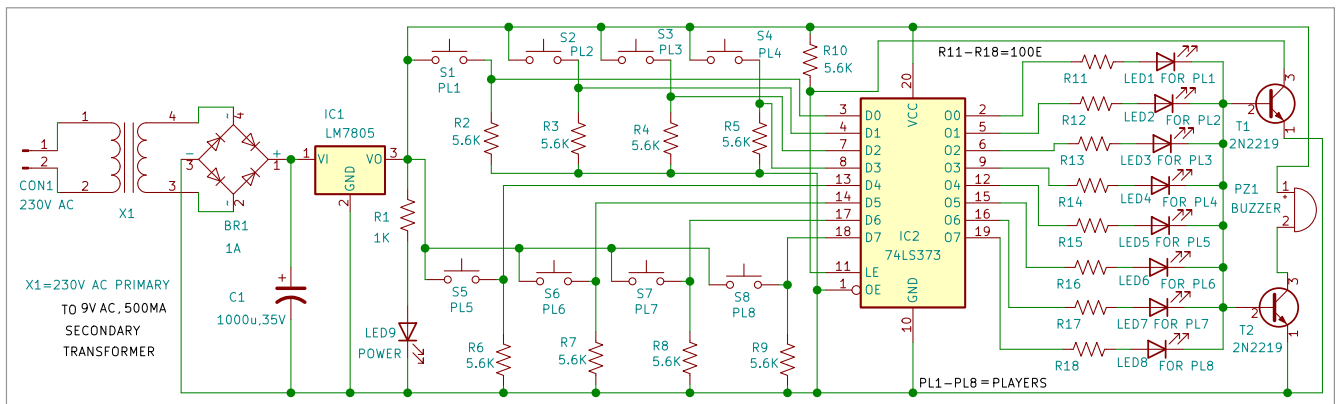


Fig. 2: Circuit diagram